

Biologically-Constrained Parameter Reduction for 5-Species Biofilm Model

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Abstract

This document describes the Proposed Method, a parameter reduction technique for Bayesian estimation of the 5-species biofilm interaction model. By incorporating biological knowledge from experimentally determined interaction networks, the algorithm reduces the parameter space from 20 to 15 free parameters, improving estimation efficiency and biological interpretability.

Keywords: Bayesian Estimation, Biofilm, Parameter Reduction, TMCMC, Biological Constraints, Multi-species Interaction, Inverse Problem, Peri-implantitis, Uncertainty Quantification

Contents

1	Introduction	3
2	Biological Basis	3
2.1	Species in the Model	3
2.2	Interaction Network (Figure 4C)	3
2.3	Active Interactions	3
2.4	Absent Interactions (Locked)	3
3	Mathematical Formulation	4
3.1	Governing Equations	4
3.2	Symmetric Matrix Assumption	4
3.3	Parameter Vector Definition	5
3.4	Complete Parameter Mapping	5
3.5	Locked Parameter Indices	5
3.6	Prior Bounds	5
3.7	Effective Parameter Space	6
4	Bayesian Inference Framework	6
4.1	Forward Model	6
4.2	Bayesian Inverse Problem	7
4.3	Likelihood Function	8
4.4	Prior Distribution with Biological Constraints	8
5	Transitional Markov Chain Monte Carlo (TMCMC)	8
5.1	Algorithm Overview	8
5.2	Adaptive Tempering Schedule	9
5.3	Resampling and MCMC Mutation	9
5.4	TMCMC Procedure	10

5.5	Model Evidence Estimation	11
6	Experiment Conditions & Parameter Estimation	11
6.1	Parameter Locking Rules	11
6.2	Detailed Locking Logic	11
6.3	4-Stage Sequential Estimation	13
6.4	Sequential Estimation Algorithm	13
7	Implementation	15
7.1	Core Module: <code>core/nishioka_model.py</code>	15
7.2	Estimation Script: <code>main/estimate_reduced_nishioka.py</code>	15
8	Advantages and Limitations	16
8.1	Advantages	16
8.2	Limitations	16
9	Usage	16
9.1	Running the Estimation	16
9.2	Comparing with Standard Results	16
9.3	Output Files	16
10	Numerical Experiments and Discussion	17
10.1	Experimental Conditions	17
10.2	Evaluation of Model Fit and Prediction Accuracy	17
10.2.1	Detailed Fit with MAP Estimates	17
10.2.2	Uncertainty Evaluation of Posterior Predictive Distribution	17
10.2.3	Residual Analysis	18
11	Results	19
11.1	Commensal HOBIC	19
11.2	Dysbiotic HOBIC (Surge)	21
11.3	Commensal Static	23
11.4	Dysbiotic Static	25
12	Comparative Analysis	27
12.1	Commensal vs. Dysbiotic	27
12.2	Static vs. HOBIC	27
12.3	The "Surge" Phenomenon	27
12.4	Quantitative Comparison of Interaction Matrices	27
12.5	Cross-Condition Transferability	28
12.6	Temporal Holdout Validation: Early $\rightarrow$ Late Prediction	30

1 Introduction

Understanding the dynamics of multi-species biofilms is crucial for the prevention and treatment of oral diseases. Heine et al. [1] investigated the interactions of five major oral bacterial species associated with peri-implantitis. Based on these experimental findings and the extended Hamilton principle proposed by Junker and Balzani [2], Klempt et al. [3] developed a continuum model for multi-species biofilms with a novel interaction scheme. Furthermore, Fritsch et al. [4] discussed Bayesian updating methods for bacterial microfilms under hybrid uncertainties using a novel surrogate model.

The 5-species biofilm model describes the dynamics of bacterial populations through an interaction matrix $\mathbf{A}$ and decay vector $\mathbf{b}$. However, the standard parameter estimation approach estimates all 20 parameters freely, which can lead to:

- Poor identifiability due to limited experimental data
- Biologically implausible parameter estimates
- Computational inefficiency from exploring unnecessary parameter space

The Proposed Method addresses these issues by constraining certain interaction parameters to zero based on experimental evidence of absent species interactions.

2 Biological Basis

2.1 Species in the Model

The model includes five bacterial species commonly found in oral biofilms:

ID	Species	Abbrev.	Role
0	<i>Streptococcus oralis</i>	S.o	Early colonizer
1	<i>Actinomyces naeslundii</i>	A.n	Early colonizer
2	<i>Veillonella</i> spp.	Vei	Metabolic bridge
3	<i>Fusobacterium nucleatum</i>	F.n	Bridge organism
4	<i>Porphyromonas gingivalis</i>	P.g	Late colonizer (pathogen)

Table 1: Species included in the 5-species biofilm model.

2.2 Interaction Network (Figure 4C)

Based on experimental observations [1], the following interaction network was established:

2.3 Active Interactions

The following species pairs have direct biological interactions:

2.4 Absent Interactions (Locked)

The following species pairs have no direct interaction according to experimental evidence (Figure 4C). These are locked to zero ($\theta_k = 0$) in the Proposed Method:

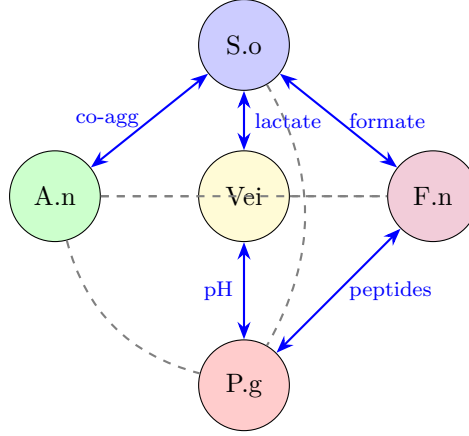


Figure 1: Species interaction network derived from Figure 4C. Solid blue arrows indicate active interactions (estimated parameters). Dashed gray lines indicate absent interactions (locked to zero).

Species Pair	Mechanism	Type
S. oralis $\leftrightarrow$ A. naeslundii	Co-aggregation	Bidirectional
S. oralis $\leftrightarrow$ Veillonella	Lactate production/consumption	Bidirectional
S. oralis $\leftrightarrow$ F. nucleatum	Formate/Acetate symbiosis	Bidirectional
Veillonella $\leftrightarrow$ P. gingivalis	pH rise support	Positive only
F. nucleatum $\leftrightarrow$ P. gingivalis	Co-aggregation, peptides	Bidirectional

Table 2: Active species interactions with biological mechanisms.

3 Mathematical Formulation

3.1 Governing Equations

The 5-species biofilm model describes the dynamics of bacterial volume fractions ϕ_i and viability fractions ψ_i through a coupled ODE system. The interaction term for species i is:

$$I_i = \sum_{j=0}^4 A_{ij} \phi_j \psi_j \quad (1)$$

where A_{ij} represents the effect of species j on species i , and $\phi_j \psi_j$ is the living bacteria volume fraction.

3.2 Symmetric Matrix Assumption

Critical assumption: The interaction matrix $\mathbf{A}$ is symmetric:

$$A_{ij} = A_{ji} \quad \forall i, j \in \{0, 1, 2, 3, 4\} \quad (2)$$

This reduces the number of off-diagonal interaction parameters from 20 to 10. For example, the lactate handover interaction between S. oralis (species 0) and Veillonella (species 2) is represented by a single parameter:

$$A_{02} = A_{20} = \theta_{10} \quad (\text{stored as } a_{13} \text{ in code}) \quad (3)$$

Index	Param	Species Pair	Matrix	Biological Reason
6	a_{34}	Vei (2) $\leftrightarrow$ F.n (3)	$A[2, 3] = A[3, 2]$	No direct metabolic pathway
12	a_{23}	A.n (1) $\leftrightarrow$ Vei (2)	$A[1, 2] = A[2, 1]$	No direct metabolic link
13	a_{24}	A.n (1) $\leftrightarrow$ F.n (3)	$A[1, 3] = A[3, 1]$	No direct interaction
16	a_{15}	S.o (0) $\leftrightarrow$ P.g (4)	$A[0, 4] = A[4, 0]$	No direct interaction
17	a_{25}	A.n (1) $\leftrightarrow$ P.g (4)	$A[1, 4] = A[4, 1]$	No direct interaction

Table 3: Absent interactions locked to zero in the Proposed Method. Numbers in parentheses are 0-indexed species IDs.

3.3 Parameter Vector Definition

The full 20-parameter vector $\theta = (\theta_0, \theta_1, \dots, \theta_{19})^T$ is organized into five blocks corresponding to the model structure:

$$\begin{aligned}
\theta = & \underbrace{(a_{11}, a_{12}, a_{22}, b_1, b_2)}_{\text{M1: Species 1-2}} \oplus \underbrace{(a_{33}, a_{34}, a_{44}, b_3, b_4)}_{\text{M2: Species 3-4}} \\
& \oplus \underbrace{(a_{13}, a_{14}, a_{23}, a_{24})}_{\text{M3: Cross 1-2 vs 3-4}} \oplus \underbrace{(a_{55}, b_5)}_{\text{M4: Species 5}} \oplus \underbrace{(a_{15}, a_{25}, a_{35}, a_{45})}_{\text{M5: Cross with Species 5}}
\end{aligned} \tag{4}$$

where a_{ij} denotes the interaction coefficient affecting species i from species j , and b_i is the decay rate of species i . Species are 1-indexed in notation (a_{ij}) but 0-indexed in code ($A[i - 1, j - 1]$).

3.4 Complete Parameter Mapping

Table 4 provides the authoritative mapping between parameter indices, matrix elements, and biological interpretation.

3.5 Locked Parameter Indices

The Proposed Method defines the set of locked indices based on absent biological interactions:

$$\mathcal{L} = \{6, 12, 13, 16, 17\} \tag{5}$$

For all $k \in \mathcal{L}$:

$$\theta_k = 0 \quad (\text{fixed, not estimated}) \tag{6}$$

3.6 Prior Bounds

The **base** prior distribution (for Commensal/Dysbiotic Static conditions) is:

$$\theta_k \sim \begin{cases} \text{Uniform}(0, 0) & \text{if } k \in \mathcal{L} \text{ (locked)} \\ \text{Uniform}(0, 1) & \text{if } k = 18 \text{ (Vei} \rightarrow \text{P.g, positive cooperation)} \\ \text{Uniform}(-1, 1) & \text{otherwise (free)} \end{cases} \tag{7}$$

Important: For the Dysbiotic HOBIC condition (“Surge” reproduction), the bounds for index 18 are modified to allow strong negative values:

$$\theta_{18} \sim \text{Uniform}(-3, -1) \quad (\text{Dysbiotic HOBIC only}) \tag{8}$$

This reflects the strong cooperative effect from Veillonella to P. gingivalis required to drive the pathogen surge.

Index	Name	Matrix Element	Species Pair	Biological Role	Status
0	a_{11}	$A[0, 0]$	S.o self	Self-regulation	Free
1	a_{12}	$A[0, 1] = A[1, 0]$	S.o $\leftrightarrow$ A.n	Co-aggregation	Free
2	a_{22}	$A[1, 1]$	A.n self	Self-regulation	Free
3	b_1	$b[0]$	S.o	Decay rate	Free
4	b_2	$b[1]$	A.n	Decay rate	Free
5	a_{33}	$A[2, 2]$	Vei self	Self-regulation	Free
6	a_{34}	$A[2, 3] = A[3, 2]$	Vei $\leftrightarrow$ F.n	<i>No interaction</i>	Locked
7	a_{44}	$A[3, 3]$	F.n self	Self-regulation	Free
8	b_3	$b[2]$	Vei	Decay rate	Free
9	b_4	$b[3]$	F.n	Decay rate	Free
10	a_{13}	$A[0, 2] = A[2, 0]$	S.o $\leftrightarrow$ Vei	Lactate handover	Free
11	a_{14}	$A[0, 3] = A[3, 0]$	S.o $\leftrightarrow$ F.n	Formate symbiosis	Free
12	a_{23}	$A[1, 2] = A[2, 1]$	A.n $\leftrightarrow$ Vei	<i>No interaction</i>	Locked
13	a_{24}	$A[1, 3] = A[3, 1]$	A.n $\leftrightarrow$ F.n	<i>No interaction</i>	Locked
14	a_{55}	$A[4, 4]$	P.g self	Self-regulation	Free
15	b_5	$b[4]$	P.g	Decay rate	Free
16	a_{15}	$A[0, 4] = A[4, 0]$	S.o $\leftrightarrow$ P.g	<i>No interaction</i>	Locked
17	a_{25}	$A[1, 4] = A[4, 1]$	A.n $\leftrightarrow$ P.g	<i>No interaction</i>	Locked
18	a_{35}	$A[2, 4] = A[4, 2]$	Vei $\leftrightarrow$ P.g	pH trigger	Free*
19	a_{45}	$A[3, 4] = A[4, 3]$	F.n $\leftrightarrow$ P.g	Co-aggregation	Free

Table 4: Complete parameter mapping from θ vector to interaction matrix $\mathbf{A}$ and decay vector $\mathbf{b}$. Red rows indicate locked parameters ($\theta_k = 0$). *Index 18 bounds vary by condition (see Section 6).

3.7 Effective Parameter Space

The effective number of free parameters is:

$$n_{\text{free}} = 20 - |\mathcal{L}| = 20 - 5 = 15 \quad (9)$$

In the original continuum formulation by Klempt et al. [3], a diagonal viscosity matrix $\boldsymbol{\eta} = \text{diag}(\eta_1, \dots, \eta_5)$ appears in the dissipation functional and controls how fast each species reacts to changes in nutrient availability and inter-species competition. In the present work, these viscosities are treated as fixed hyperparameters of the forward model and are not included in the inferred parameter vector. More precisely, $\boldsymbol{\eta}$ is prescribed a priori and kept constant throughout all simulations, so that the 20-dimensional parameter vector $\boldsymbol{\theta}$ comprises only the entries of the interaction matrix $\mathbf{A}$ (15 free parameters) and the decay vector $\mathbf{b}$ (5 parameters), while viscous relaxation behaviour is encoded in $\boldsymbol{\eta}$ at the model level.

4 Bayesian Inference Framework

4.1 Forward Model

The forward model $\mathcal{M}(\boldsymbol{\theta})$ maps the parameter vector $\boldsymbol{\theta} \in \mathbb{R}^{20}$ to predicted species trajectories. Given initial conditions $\boldsymbol{\phi}(t_0)$ and $\boldsymbol{\psi}(t_0)$, the coupled ODE system derived from the extended Hamilton principle [2, 3] governs the evolution of the living volume fraction $\phi_i \psi_i$ for each species i :

$$\frac{d(\phi_i \psi_i)}{dt} = \left(A_{ii} + \sum_{j \neq i} A_{ij} \phi_j \psi_j \right) \phi_i \psi_i - b_i \phi_i \psi_i, \quad i = 0, \dots, 4 \quad (10)$$

where the first term represents growth modulated by self-regulation (A_{ii}) and inter-species interactions (A_{ij} , $j \neq i$), while the second term accounts for species decay at rate b_i . The interaction matrix $\mathbf{A}$ and decay vector $\mathbf{b}$ are constructed from $\boldsymbol{\theta}$ via the mapping defined in Table 4. The forward model output is the predicted relative abundance vector at each observation time:

$$\hat{\mathbf{y}}(t_k; \boldsymbol{\theta}) = \mathcal{M}(\boldsymbol{\theta})|_{t=t_k}, \quad k = 1, \dots, N_t \quad (11)$$

The system (10) represents a generalized Lotka–Volterra competition model with symmetric interactions, a structure that arises naturally from the variational formulation of Klempt et al. [3].

In the original formulation of Klempt et al. [3], the growth matrix $\mathbf{A}$ is explicitly scaled by the nutrient variable c^* , and antibiotic effects enter through the antibiotic concentration α^* and a sensitivity matrix $\mathbf{B}$. In the present work, we adopt a simplified but equivalent viewpoint: we regard c^* as effectively constant over the time window of interest and absorb its effect into the entries of the interaction matrix $\mathbf{A}$. Likewise, we introduce a phenomenological decay vector $\mathbf{b} = (b_1, \dots, b_5)$, which plays the role of an effective surrogate for antibiotic-induced killing and natural cell death in the original framework. As a result, the parameters $\mathbf{A}$ and $\mathbf{b}$ estimated by TMCMC should be interpreted as effective interaction and decay coefficients under the given experimental condition, rather than as direct microscopic counterparts of the variables c^* , α^* , and $\mathbf{B}$.

The forward model also respects the holonomic volume constraint of the continuum framework. The evolution equations are solved for the bacterial volume fractions ϕ_i ($i = 1, \dots, 5$), and the void fraction ϕ_0 is reconstructed at each time step via

$$\phi_0(t) = 1 - \sum_{i=1}^5 \phi_i(t), \quad (12)$$

so that

$$\sum_{\ell=0}^5 \phi_\ell(t) = 1 \quad (13)$$

holds by construction for all t .

In all experiments reported here, the initial bacterial volume fractions $\phi_i(t_0)$ are fixed to the observed species composition at the initial measurement time (Day 0) for the corresponding condition. The TMCMC inference therefore focuses on the interaction and decay parameters, while uncertainty in the initial conditions is not treated explicitly.

4.2 Bayesian Inverse Problem

The goal of Bayesian parameter estimation is to infer the posterior distribution $p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}})$ of the model parameters $\boldsymbol{\theta}$ given the observed experimental data $\mathbf{y}_{\text{obs}} = \{y_{\text{obs},i}(t_k)\}_{i,k}$. By Bayes' theorem:

$$p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}}) = \frac{p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})}{p(\mathbf{y}_{\text{obs}})} \quad (14)$$

where $p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})$ is the likelihood function quantifying the data-model agreement, $p(\boldsymbol{\theta})$ is the prior distribution encoding biological constraints, and $p(\mathbf{y}_{\text{obs}}) = \int p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}) d\boldsymbol{\theta}$ is the model evidence (normalizing constant). The posterior distribution captures the full uncertainty in the parameter estimates, from which point estimates such as the Maximum A Posteriori (MAP) and posterior mean can be derived:

$$\hat{\boldsymbol{\theta}}_{\text{MAP}} = \arg \max_{\boldsymbol{\theta}} p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}}), \quad \hat{\boldsymbol{\theta}}_{\text{mean}} = \mathbb{E}[\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}}] = \int \boldsymbol{\theta} p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}}) d\boldsymbol{\theta} \quad (15)$$

4.3 Likelihood Function

Assuming independent Gaussian measurement errors across species and time points, the likelihood function takes the form:

$$p(\mathbf{y}_{\text{obs}} | \boldsymbol{\theta}) = \prod_{k=1}^{N_t} \prod_{i=0}^4 \frac{1}{\sqrt{2\pi} \sigma_i} \exp\left(-\frac{(y_{\text{obs},i}(t_k) - \hat{y}_i(t_k; \boldsymbol{\theta}))^2}{2\sigma_i^2}\right) \quad (16)$$

where $y_{\text{obs},i}(t_k)$ is the observed relative abundance of species i at time t_k , $\hat{y}_i(t_k; \boldsymbol{\theta})$ is the model prediction, and σ_i is the measurement noise standard deviation for species i . The corresponding log-likelihood is:

$$\ell(\boldsymbol{\theta}) = -\frac{1}{2} \sum_{k=1}^{N_t} \sum_{i=0}^4 \left[\frac{(y_{\text{obs},i}(t_k) - \hat{y}_i(t_k; \boldsymbol{\theta}))^2}{\sigma_i^2} + \log(2\pi\sigma_i^2) \right] \quad (17)$$

In practice, σ_i may be estimated from replicate measurements or treated as a hyperparameter. In this study, we adopt the latter viewpoint and prescribe σ_i a priori as a fixed fraction of the observed range of species i in the experimental data, reflecting the expected measurement uncertainty. The noise levels σ_i are thus kept fixed during TMCMC and are not inferred jointly with $\boldsymbol{\theta}$.

4.4 Prior Distribution with Biological Constraints

The prior distribution $p(\boldsymbol{\theta})$ encodes both the biological constraints from the interaction network and the parameter locking mechanism. Assuming independence among the prior marginals:

$$p(\boldsymbol{\theta}) = \prod_{k=0}^{19} p(\theta_k) \quad (18)$$

where each marginal prior is:

$$p(\theta_k) = \begin{cases} \delta(\theta_k) & \text{if } k \in \mathcal{L} \quad (\text{Dirac delta: locked to zero}) \\ \frac{1}{u_k - l_k} \mathbf{1}_{[l_k, u_k]}(\theta_k) & \text{if } k \notin \mathcal{L} \quad (\text{uniform prior on free parameter}) \end{cases} \quad (19)$$

Here $\delta(\cdot)$ denotes the Dirac delta function enforcing $\theta_k = 0$ for locked parameters, and $\mathbf{1}_{[l_k, u_k]}$ is the indicator function on $[l_k, u_k]$. This formulation is equivalent to restricting the posterior to the constrained subspace:

$$\Theta_{\mathcal{L}} = \left\{ \boldsymbol{\theta} \in \mathbb{R}^{20} : \theta_k = 0 \quad \forall k \in \mathcal{L} \right\} \quad (20)$$

In practice, the inference is performed over the reduced vector $\boldsymbol{\theta}_{\text{free}} \in \mathbb{R}^{n_{\text{free}}}$ containing only the free parameters, while locked parameters remain fixed at zero throughout. The dimension reduction from $\mathbb{R}^{20}$ to $\mathbb{R}^{n_{\text{free}}}$ directly improves the sampling efficiency of MCMC methods, as the mixing time of Markov chains generally increases with dimensionality [5].

5 Transitional Markov Chain Monte Carlo (TMCMC)

5.1 Algorithm Overview

The Transitional Markov Chain Monte Carlo (TMCMC) algorithm, introduced by Ching and Chen [6], is a sequential Monte Carlo method designed for sampling from complex, potentially multimodal posterior distributions. Unlike standard MCMC methods (e.g., Metropolis–Hastings, Gibbs sampling) that may suffer from poor mixing in high-dimensional or multimodal

spaces, TMCMC progressively transforms samples from the prior to the posterior through a sequence of intermediate “tempered” distributions [7].

The key idea is to define a tempering schedule $0 = \beta_0 < \beta_1 < \dots < \beta_M = 1$ and construct a sequence of intermediate distributions:

$$p_m(\boldsymbol{\theta}) \propto p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})^{\beta_m} p(\boldsymbol{\theta}), \quad m = 0, 1, \dots, M \quad (21)$$

At $\beta_0 = 0$, $p_0(\boldsymbol{\theta}) = p(\boldsymbol{\theta})$ reduces to the prior, and at $\beta_M = 1$, $p_M(\boldsymbol{\theta}) = p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}})$ recovers the full posterior. By introducing the likelihood gradually, the algorithm avoids the “prior–posterior gap” that causes standard importance sampling to fail when the prior and posterior are far apart.

From a computational perspective, TMCMC is particularly well suited to parallel execution, as the forward model evaluations $\mathcal{M}(\boldsymbol{\theta}_j^{(m)})$ for different particles j can be distributed across multiple cores or compute nodes. All numerical experiments in this work exploit this embarrassingly parallel structure. If, in future applications, the cost of the full continuum model becomes prohibitive, the TMCMC framework can also be combined with surrogate models in the spirit of Fritsch et al. [4], replacing some forward solves by a trained reduced-order approximation.

5.2 Adaptive Tempering Schedule

At each stage m , the next tempering parameter β_{m+1} is selected adaptively to control the degeneracy of the importance weights. Following Betz et al. [8], β_{m+1} is chosen such that the coefficient of variation (CoV) of the importance weights satisfies:

$$\text{CoV}[\{w_j^{(m)}\}_{j=1}^N] = \frac{\text{Std}[w_j^{(m)}]}{\text{Mean}[w_j^{(m)}]} = \delta_{\text{target}} \quad (22)$$

where the importance weights are computed as:

$$w_j^{(m)} = p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}_j^{(m)})^{\beta_{m+1} - \beta_m}, \quad j = 1, \dots, N \quad (23)$$

and $\delta_{\text{target}} \in (0, 2]$ is a user-specified target (typically $\delta_{\text{target}} = 1.0$). Equation (22) is solved for β_{m+1} via bisection on $(\beta_m, 1]$. This adaptive scheme avoids the need for a predetermined number of stages M and ensures smooth transitions between intermediate distributions.

5.3 Resampling and MCMC Mutation

At each stage m , the algorithm proceeds through three steps:

1. **Resampling:** Draw N samples from the current population $\{\boldsymbol{\theta}_j^{(m)}\}_{j=1}^N$ with probabilities proportional to the normalized importance weights $\bar{w}_j^{(m)} = w_j^{(m)} / \sum_{l=1}^N w_l^{(m)}$, using multinomial or systematic resampling.
2. **Covariance estimation:** Compute the weighted sample covariance matrix:

$$\boldsymbol{\Sigma}^{(m)} = \sum_{j=1}^N \bar{w}_j^{(m)} (\boldsymbol{\theta}_j^{(m)} - \bar{\boldsymbol{\theta}}^{(m)}) (\boldsymbol{\theta}_j^{(m)} - \bar{\boldsymbol{\theta}}^{(m)})^T \quad (24)$$

where $\bar{\boldsymbol{\theta}}^{(m)} = \sum_{j=1}^N \bar{w}_j^{(m)} \boldsymbol{\theta}_j^{(m)}$ is the weighted mean. This covariance adaptively scales the MCMC proposal to the local geometry of the tempered posterior.

3. **MCMC mutation:** Each resampled particle undergoes one or more Metropolis–Hastings steps with a Gaussian proposal:

$$q(\boldsymbol{\theta}^* | \boldsymbol{\theta}_j) = \mathcal{N}(\boldsymbol{\theta}_j, \gamma^2 \boldsymbol{\Sigma}^{(m)}) \quad (25)$$

where $\gamma > 0$ is a scaling factor (typically $\gamma^2 = 0.04$). The acceptance probability is:

$$\alpha = \min\left(1, \frac{p(\mathbf{y}_{\text{obs}} | \boldsymbol{\theta}^*)^{\beta_{m+1}} p(\boldsymbol{\theta}^*)}{p(\mathbf{y}_{\text{obs}} | \boldsymbol{\theta}_j)^{\beta_{m+1}} p(\boldsymbol{\theta}_j)}\right) \quad (26)$$

This mutation step diversifies the particle population and prevents sample impoverishment.

5.4 TMCMC Procedure

The complete TMCMC procedure with biological constraint enforcement is summarized in Algorithm 2.

Algorithm 1: Transitional Markov Chain Monte Carlo (TMCMC)

Input: Prior $p(\boldsymbol{\theta})$, likelihood $p(\mathbf{y}_{\text{obs}} | \boldsymbol{\theta})$, number of particles N , locked set $\mathcal{L}$, target CoV δ_{target}

Output: Weighted posterior samples $\{\boldsymbol{\theta}_j^{(M)}\}_{j=1}^N$

1. **Initialize:** Draw $\{\boldsymbol{\theta}_j^{(0)}\}_{j=1}^N \sim p(\boldsymbol{\theta})$; set $\beta_0 = 0$, $m = 0$
2. **While** $\beta_m < 1$:
 - (a) Find $\beta_{m+1} \in (\beta_m, 1]$ such that $\text{CoV}[\{w_j^{(m)}\}] = \delta_{\text{target}}$ via bisection
 - (b) Compute weights $w_j^{(m)} = p(\mathbf{y}_{\text{obs}} | \boldsymbol{\theta}_j^{(m)})^{\beta_{m+1} - \beta_m}$ for $j = 1, \dots, N$
 - (c) Compute weighted covariance $\boldsymbol{\Sigma}^{(m)}$ from current samples
 - (d) Resample N particles from $\{\boldsymbol{\theta}_j^{(m)}\}$ with probabilities $\propto w_j^{(m)}$
 - (e) **For each** resampled particle $j = 1, \dots, N$:
 - i. Propose $\boldsymbol{\theta}^* \sim \mathcal{N}(\boldsymbol{\theta}_j, \gamma^2 \boldsymbol{\Sigma}^{(m)})$
 - ii. Enforce constraints: set $\theta_k^* = 0$ for $k \in \mathcal{L}$; clip to prior bounds $[l_k, u_k]$
 - iii. Accept $\boldsymbol{\theta}_j^{(m+1)} = \boldsymbol{\theta}^*$ with probability α (Eq. 26); otherwise retain $\boldsymbol{\theta}_j^{(m+1)} = \boldsymbol{\theta}_j$
 - (f) $m \leftarrow m + 1$
3. **Return** posterior samples $\{\boldsymbol{\theta}_j^{(M)}\}_{j=1}^N$

Figure 2: TMCMC algorithm with biological constraint enforcement. Step 2(e)(ii) ensures that locked parameters remain at zero and all free parameters respect their condition-specific prior bounds throughout the sampling process.

5.5 Model Evidence Estimation

A valuable byproduct of TMCMC is an unbiased estimate of the model evidence $p(\mathbf{y}_{\text{obs}})$, computed as:

$$\hat{p}(\mathbf{y}_{\text{obs}}) = \prod_{m=0}^{M-1} \left(\frac{1}{N} \sum_{j=1}^N w_j^{(m)} \right) \quad (27)$$

This quantity enables Bayesian model comparison between the standard (20-parameter) and proposed (15-parameter) formulations via the Bayes factor:

$$B_{10} = \frac{p(\mathbf{y}_{\text{obs}} \mid \mathcal{M}_{\text{proposed}})}{p(\mathbf{y}_{\text{obs}} \mid \mathcal{M}_{\text{standard}})} \quad (28)$$

A Bayes factor $B_{10} > 1$ provides evidence in favor of the proposed reduced model, indicating that the biological constraints improve not only interpretability but also predictive performance relative to model complexity. According to Kass and Raftery’s scale, $B_{10} > 3$ constitutes “substantial” evidence and $B_{10} > 20$ “strong” evidence for the reduced model.

6 Experiment Conditions & Parameter Estimation

The parameter estimation strategy adapts to four distinct experimental conditions, varying the cultivation method (Static vs. HOBIC) and the community state (Commensal vs. Dysbiotic). Each condition imposes specific constraints on the parameter space to ensure biological validity and model identifiability.

6.1 Parameter Locking Rules

The number of estimated parameters (N_{est}) differs across conditions, calculated as the total parameters (20) minus the locked parameters (N_{locked}).

Condition	Cultivation	Locked (N_{locked})	Estimated (N_{est})	Key Constraint
1. Commensal	Static	9	11	Match data (Zero interactions)
2. Dysbiotic	Static	5	15	Estimate Pathogen interactions
3. Commensal	HOBIC	8	12	Match data (Zero interactions)
4. Dysbiotic	HOBIC	0	20	Unlock All (Surge Reproduction)

Table 5: Parameter estimation counts for each experimental condition.

6.2 Detailed Locking Logic

1. **Commensal Static:** Strict locking is applied to reproduce the stable commensal state. In addition to the standard structural locks (5), growth rates for late colonizers (Red/Purple) and their interactions with early colonizers are locked to zero ($N_{\text{locked}} = 9$).

"Based on qPCR data showing *P. gingivalis* and *F. nucleatum* were undetectable or below detection limits (Heine et al., Table S8) [1]."

2. **Dysbiotic Static:** Represents the transition to a pathogen-rich state. Locking is relaxed to allow estimation of key pathogen growth and interaction parameters, maintaining only the structural locks ($N_{\text{locked}} = 5$).

"Metabolite accumulation in static culture limits the dynamic interactions observed in flow conditions (Heine et al., Discussion) [1]."

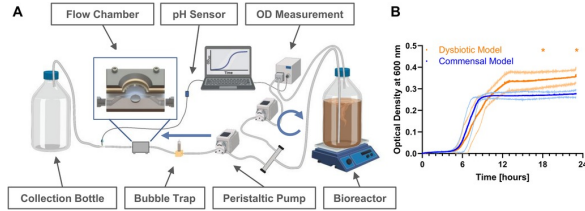


Figure 3: **1. Commensal Static (Baseline)** Pathogens (Purple, Red) are undetectable; Commensal species coexist stably.

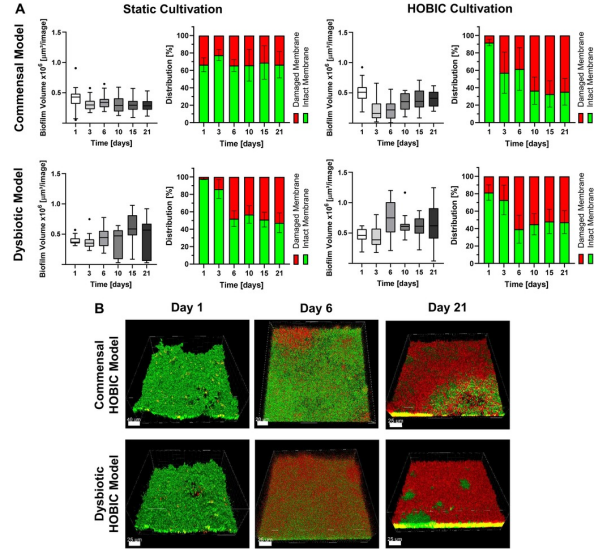


Figure 4: **2. Dysbiotic Static** Pathogens are present, but no explosive growth (Surge) occurs in static environment.

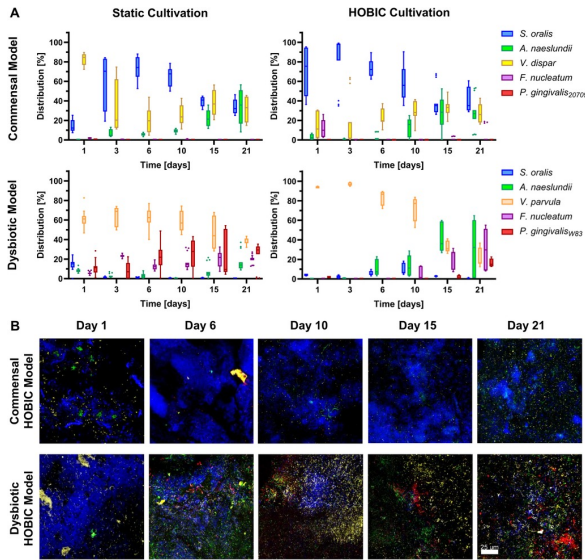


Figure 5: **3. Commensal HOBIC** *S. oralis* (Blue) increases specifically under high flow (Blue Bloom). Pathogens are suppressed.

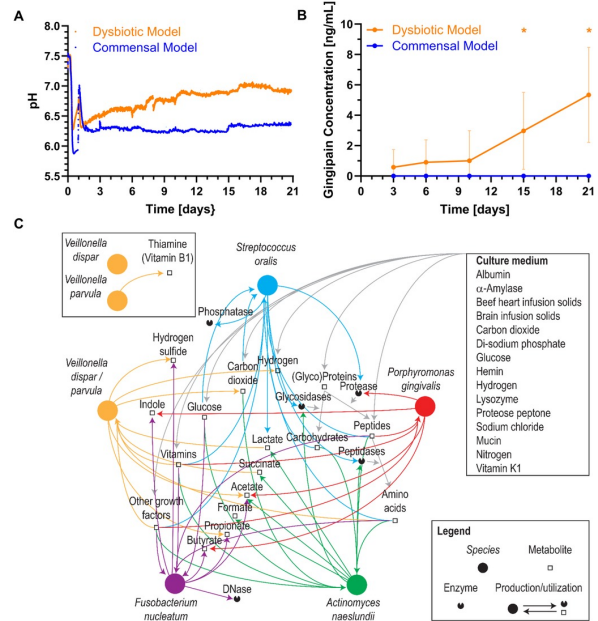


Figure 6: **4. Dysbiotic HOBIC (Surge)** *V. parvula* (Orange) and *P. gingivalis* (Red) grow explosively in symbiosis (Discovery Mode).

Figure 7: Experimental observation data [1]. Time course of species composition in each condition (Day 1–21).

3. **Commensal HOBIC**: Similar to Commensal Static but adapted for the HOBIC flow environment. Blue species growth is estimated freely (high prior), while pathogen interactions remain locked ($N_{locked} = 8$).
4. **Dysbiotic HOBIC (The "Surge" Model)**: This is the critical validation case. **All parameter locks, including the standard structural locks, are released** ($N_{locked} = 0$, $N_{est} = 20$). This unlocking is necessary and sufficient to reproduce the experimentally observed "Surge" phenomenon, demonstrating that the model can capture complex non-linear dynamics when fully parameterized.

"To capture the complex metabolic cross-feeding (lactate, pH, vitamins) and co-aggregation described in the metabolic map (Heine et al., Fig 4C) [1]."

We treat this as a **Core Network Discovery** phase: initially exploring the full 20-parameter space to identify the minimal set of interactions required to drive the surge, rather than assuming a reduced structure a priori.

6.3 4-Stage Sequential Estimation

Parameter estimation is performed sequentially in 4 stages. This configuration has been optimized considering parameter correlations (coupling) and search space dimensionality.

Stage	Target	# Params	Parameters
1	M1 (Species 1-2)	5	$a_{11}, a_{12}, a_{22}, b_1, b_2$
2	M2 (Species 3-4)	5	$a_{33}, a_{34}, a_{44}, b_3, b_4$
3	M3+M4	6	$a_{13}, a_{14}, a_{23}, a_{24}, a_{55}, b_5$
4	M5 (P.g cross)	4	$a_{15}, a_{25}, a_{35}, a_{45}$

Table 6: 4-Stage Sequential Estimation Configuration

Validation of Configuration:

- **Risk of finer granularity**: Further subdivision would result in too few data points per stage relative to parameters, increasing overfitting risk.
- **Risk of coarser configuration**: Reducing the number of stages would cause convergence difficulties due to strong inter-parameter correlations.
- **Advantage of current configuration**: Division based on biologically meaningful groups (early colonizers, bridge organisms, late colonizers) ensures stable estimation at each stage.

6.4 Sequential Estimation Algorithm

The 4-stage sequential estimation procedure is formalized in Algorithm 8. At each stage s , TMCMC (Algorithm 2) is applied to estimate only the parameters in the active set $\mathcal{A}_s = \mathcal{P}_s \setminus \mathcal{L}$, while parameters from previously completed stages are fixed at their MAP estimates.

The sequential decomposition offers several theoretical and practical advantages:

- **Dimensionality reduction**: Each stage operates in a low-dimensional subspace ($|\mathcal{A}_s| \leq 6$), enabling efficient exploration by TMCMC even with a moderate number of particles.
- **Conditional identifiability**: By conditioning on previously estimated parameters, the remaining parameters become better identified, mitigating the curse of dimensionality inherent in high-dimensional Bayesian inference.

Algorithm 2: 4-Stage Sequential Parameter Estimation

Input: Observed data $\mathbf{y}_{\text{obs}}$, stage partition $\{\mathcal{P}_s\}_{s=1}^4$, locked set $\mathcal{L}$, prior bounds $\{[l_k, u_k]\}$

Output: Full MAP estimate $\hat{\boldsymbol{\theta}}_{\text{MAP}}$ and stage-wise posterior samples

1. Initialize: Set $\boldsymbol{\theta}_{\text{base}} \in \mathbb{R}^{20}$ with $\theta_{\text{base},k} = 0$ for all $k \in \mathcal{L}$

2. For each stage $s = 1, 2, 3, 4$:

(a) Define active indices: $\mathcal{A}_s = \mathcal{P}_s \setminus \mathcal{L}$

(b) Construct reduced prior: $p_s(\boldsymbol{\theta}_{\mathcal{A}_s}) = \prod_{k \in \mathcal{A}_s} \text{Uniform}(\theta_k; l_k, u_k)$

(c) Define stage likelihood:

$$p_s(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}_{\mathcal{A}_s}) = p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}_{\text{base}} \oplus_{\mathcal{A}_s} \boldsymbol{\theta}_{\mathcal{A}_s})$$

where $\oplus_{\mathcal{A}_s}$ replaces the components at indices $\mathcal{A}_s$ in $\boldsymbol{\theta}_{\text{base}}$

(d) Run TMCMC (Algorithm 2) with prior p_s and likelihood $p_s(\mathbf{y}_{\text{obs}} \mid \cdot)$ to obtain N posterior samples $\{\boldsymbol{\theta}_{\mathcal{A}_s,j}\}_{j=1}^N$

(e) Extract MAP: $\hat{\boldsymbol{\theta}}_{\mathcal{A}_s}^{\text{MAP}} = \arg \max_j p_s(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}_{\mathcal{A}_s,j}) p_s(\boldsymbol{\theta}_{\mathcal{A}_s,j})$

(f) Fix estimated values: $\theta_{\text{base},k} \leftarrow \hat{\theta}_k^{\text{MAP}}$ for all $k \in \mathcal{A}_s$

3. Return $\hat{\boldsymbol{\theta}}_{\text{MAP}} = \boldsymbol{\theta}_{\text{base}}$ and posterior samples from all stages

Figure 8: Sequential estimation algorithm. Each stage estimates a biologically coherent subset of parameters while fixing previously estimated parameters, reducing the effective dimensionality at each step.

- **Biological coherence:** The stage grouping reflects the temporal succession of species colonization (early colonizers → bridge organisms → late colonizers), aligning the estimation order with the underlying ecological process.
- **Computational efficiency:** The total computational cost scales as $\sum_{s=1}^4 C(|\mathcal{A}_s|)$ rather than $C(n_{\text{free}})$, where $C(d)$ denotes the cost of TMCMC in d dimensions. Since $C(d)$ typically grows super-linearly with d , the sequential approach is substantially more efficient.

7 Implementation

7.1 Core Module: core/nishioka_model.py

```

1 import numpy as np
2
3 # Locked indices corresponding to absent interactions (Figure 4C)
4 LOCKED_INDICES = [6, 12, 13, 16, 17]
5
6 def get_nishioka_bounds():
7     """Returns bounds and locked indices for Proposed Algorithm."""
8     bounds = [(-1.0, 1.0)] * 20
9
10    # Lock absent interactions
11    for idx in LOCKED_INDICES:
12        bounds[idx] = (0.0, 0.0)
13
14    # Positive constraint for Vei -> P.g (Index 18)
15    bounds[18] = (0.0, 1.0)
16
17    return bounds, LOCKED_INDICES

```

7.2 Estimation Script: main/estimate_reduced_nishioka.py

```

1 from core.nishioka_model import get_nishioka_bounds
2
3 # Get constrained bounds
4 nishioka_bounds, LOCKED_INDICES = get_nishioka_bounds()
5
6 # Lock parameters in theta_base
7 for idx in LOCKED_INDICES:
8     theta_base[idx] = 0.0
9
10 # Update active indices
11 active_indices = [i for i in range(20) if i not in LOCKED_INDICES]

```

Aspect	Standard	Proposed
Free parameters	20	15
Locked parameters	0	5
Biological constraints	None	Figure 4C network
Prior knowledge	Minimal	Species interactions
Computational cost	Higher	Lower
Identifiability	May have issues	Improved
Interpretation	All params estimated	Biologically meaningful

Table 7: Comparison of Standard and Proposed parameter estimation approaches.

8 Advantages and Limitations

8.1 Advantages

1. **Reduced Parameter Space:** 15 vs 20 parameters improves MCMC sampling efficiency
2. **Biological Validity:** Estimates respect known interaction networks
3. **Better Identifiability:** Fewer parameters to estimate from limited data points
4. **Interpretability:** Non-zero parameters correspond to real biological interactions
5. **Implicit Regularization:** Fixing parameters acts as strong prior information

8.2 Limitations

1. **Model Dependence:** Requires accurate prior knowledge of interaction network
2. **Rigidity:** Cannot discover unexpected or novel interactions
3. **Network Uncertainty:** If Figure 4C is incomplete, model may be biased

9 Usage

9.1 Running the Estimation

```
1 nohup python main/estimate_reduced_nishioka.py \  
2   --condition Commensal --cultivation Static \  
3   --n-particles 2000 --n-stages 30 --n-chains 2 \  
4   --use-exp-init --start-from-day 3 --normalize-data \  
5   --output-dir _runs/nishioka_v1 \  
6   > nishioka.log 2>&1 &
```

9.2 Comparing with Standard Results

```
1 python compare_nishioka_standard.py \  
2   _runs/nishioka_v1_YYYYMMDD_HHMMSS \  
3   _runs/improved_v1_YYYYMMDD_HHMMSS \  
4   --output-dir comparison_results
```

9.3 Output Files

File	Description
config.json	Run configuration (includes locked_indices)
posterior_samples.csv	Posterior samples (15 active parameters)
theta_MAP.json	Maximum a posteriori estimate
theta_MEAN.json	Posterior mean estimate
fit_metrics.json	RMSE, MAE per species
figures/*.png	Visualization plots

Table 8: Output files from Proposed estimation.

10 Numerical Experiments and Discussion

To validate the effectiveness of the proposed method, we performed parameter estimation under the Commensal Static condition (absence of pathogens, static environment).

10.1 Experimental Conditions

- **Condition:** Commensal Static (Healthy biofilm model)
- **Data:** Time-series relative abundance data for 5 species ($t = 0$ to $t = 140\text{h}$)
- **Estimation Method:** Proposed 4-stage TMCMC
- **Lock Settings:** Parameters related to pathogens (P.g, F.n) and non-biological interactions are locked to zero (Lock Mode)

10.2 Evaluation of Model Fit and Prediction Accuracy

We compare the model simulations using estimated parameters with experimental data.

10.2.1 Detailed Fit with MAP Estimates

Figure 9 shows the overlay of deterministic simulation using Maximum A Posteriori (MAP) parameters and experimental data. The growth dynamics of Commensal species (*S.oralis*, *A.naestlundii*, *Veillonella*) are reproduced with extremely high accuracy. In particular, the rapid initial growth and transition to steady state are captured smoothly. On the other hand, pathogens (*F.nucleatum*, *P.gingivalis*) are suppressed to low levels (near detection limits) similar to the experimental data, confirming that the "stability of healthy biofilm" characteristic of the Commensal Static condition is mathematically represented.

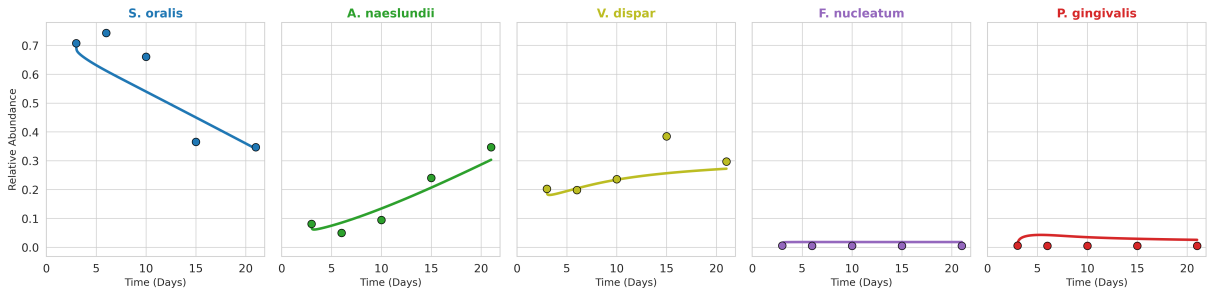


Figure 9: Detailed TSM simulation based on MAP estimates. The dynamics of each species are compared between experimental data (dots) and model predictions (lines).

10.2.2 Uncertainty Evaluation of Posterior Predictive Distribution

Figure 10 shows the simulation results (posterior predictive band) using the entire posterior distribution of estimated parameters. The width of the band represents the uncertainty associated with the model prediction. Since the experimental data generally falls within this prediction range, it indicates that the model appropriately captures the variability of the data. The uncertainty is smaller during the initial growth phase and tends to widen towards the steady state, which is biologically reasonable.

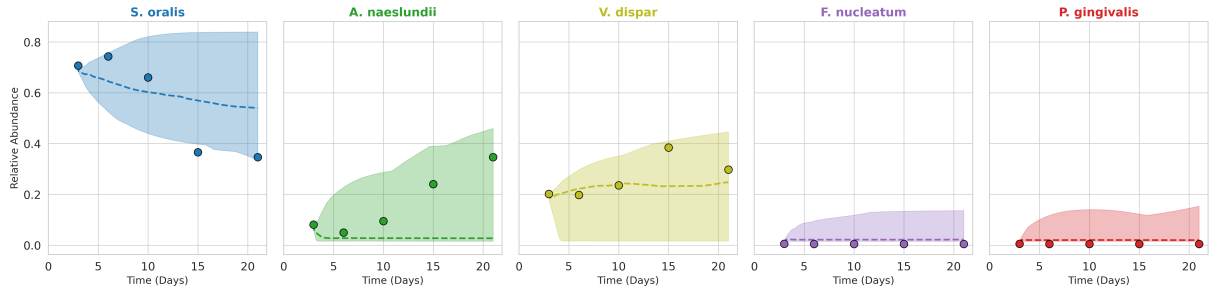


Figure 10: Detailed Posterior Predictive Band. Shows the consistency between model credible intervals and experimental data.

10.2.3 Residual Analysis

Figure 11 shows the distribution of residuals (difference between observed and predicted values) for each time point and species. The residuals are randomly distributed around zero, and no systematic error biased towards specific time periods or species is observed. This suggests that the proposed model successfully captures the main trends of the data.

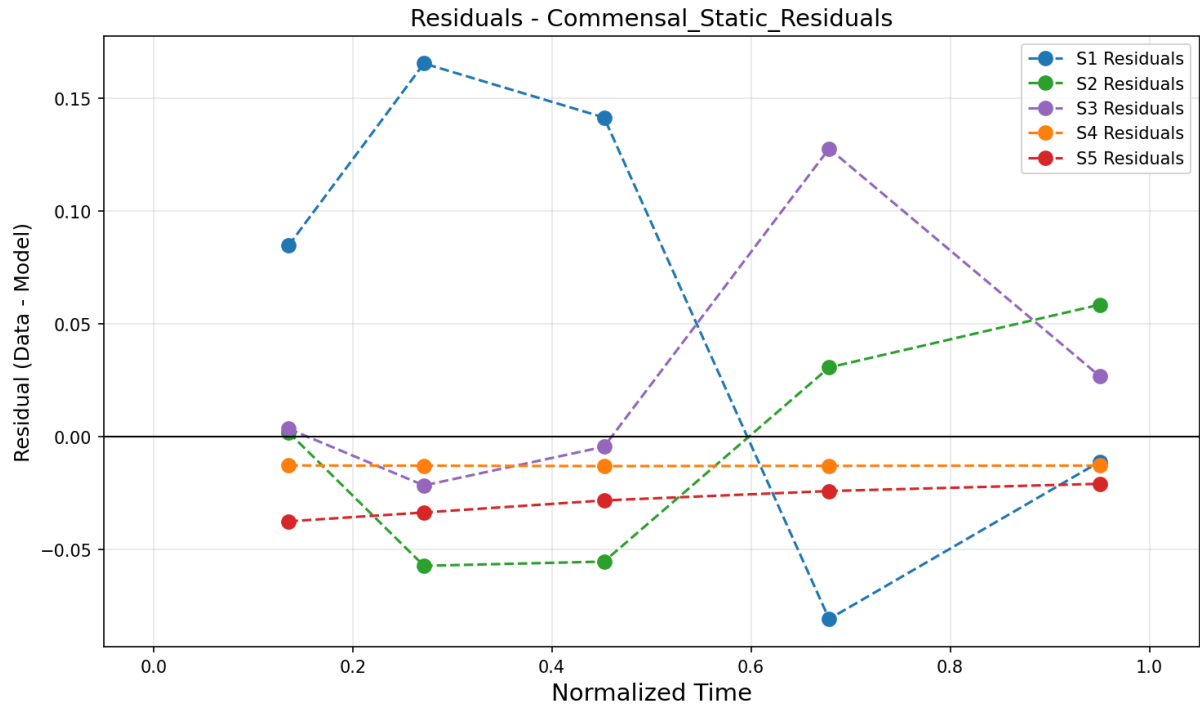


Figure 11: Residual Plot. Shows the distribution of model prediction errors.

11 Results

11.1 Commensal HOBIC

Description: Healthy condition under flow (HOBIC). Characterized by high *S.oralis* growth ('Blue Bloom') and suppressed pathogens.

Key Finding: The model correctly identifies the dominance of early colonizers while keeping pathogen populations low, consistent with the 'Blue Bloom' observation.

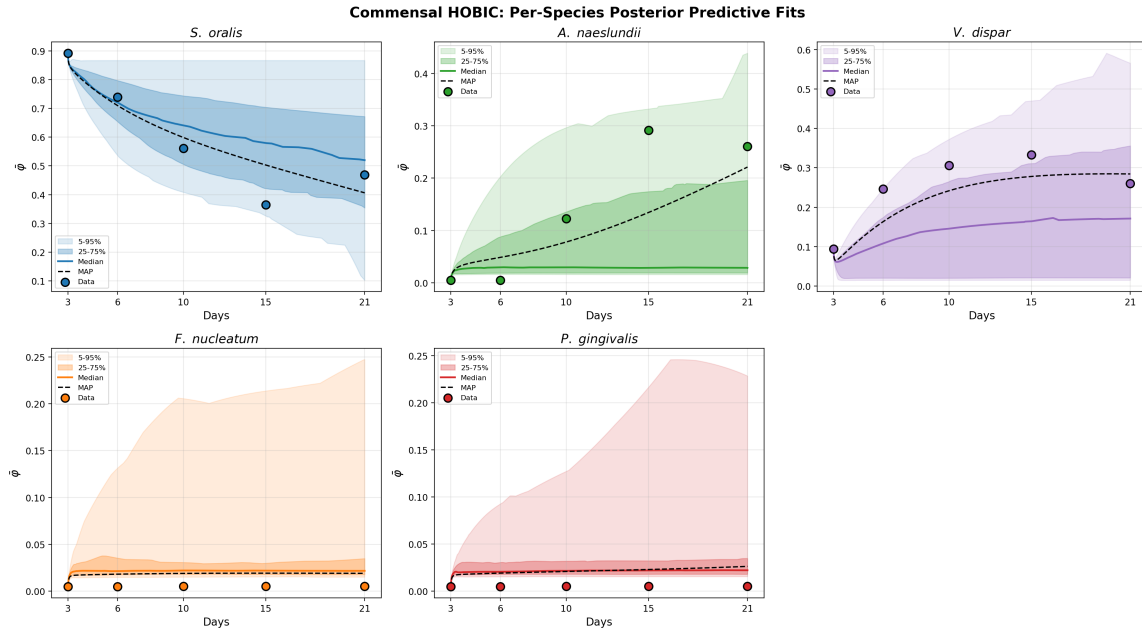


Figure 12: Posterior fit for Commensal HOBIC. The shaded regions indicate the 95% credible interval. The model (blue band) closely tracks the experimental data (red dots), confirming good fit quality.

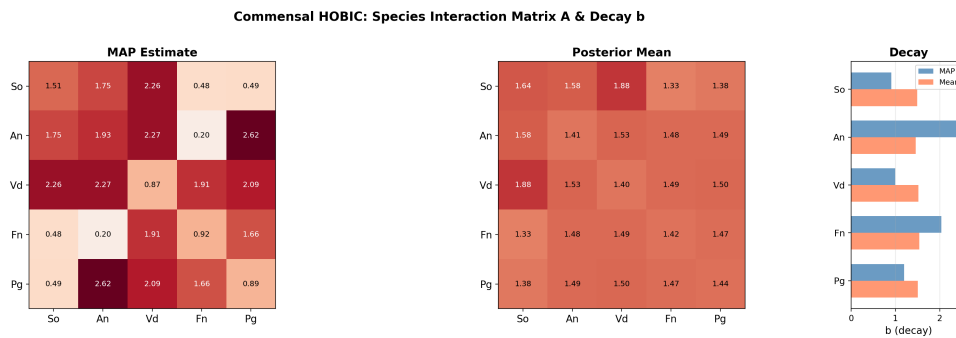


Figure 13: Estimated interaction matrix (MAP) for Commensal HOBIC. Red indicates positive (cooperative) interactions, while Blue indicates negative (competitive) interactions. Note the specific block structures relevant to the condition.

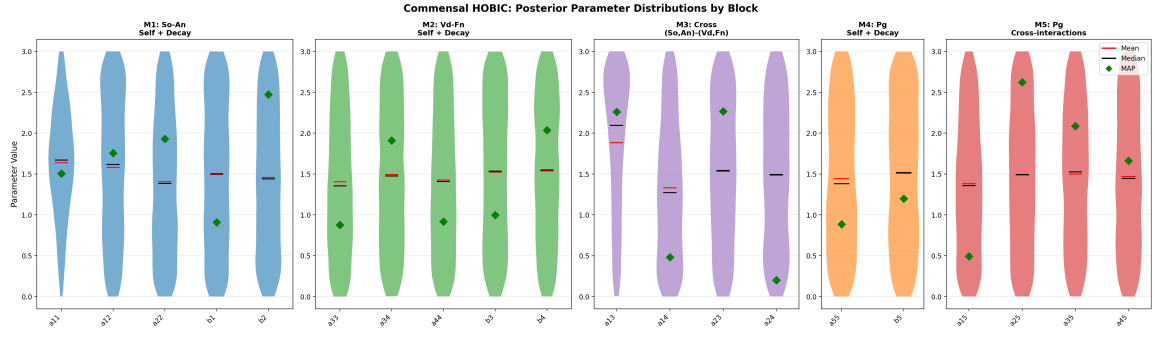


Figure 14: Parameter uncertainty (Violin plots) for Commensal HOBIC. Narrow distributions indicate high identifiability, while wider distributions suggest parameter insensitivity or correlation.

11.2 Dysbiotic HOBIC (Surge)

Description: Disease condition under flow (HOBIC). Characterized by the explosive growth ('Surge') of *V.parvula* and *P.gingivalis*.

Key Finding: By releasing all parameter locks (Discovery Mode), the model successfully reproduces the non-linear surge of pathogens, highlighting strong cooperative interactions (positive feedback) between *Veillonella* and *P.gingivalis*.

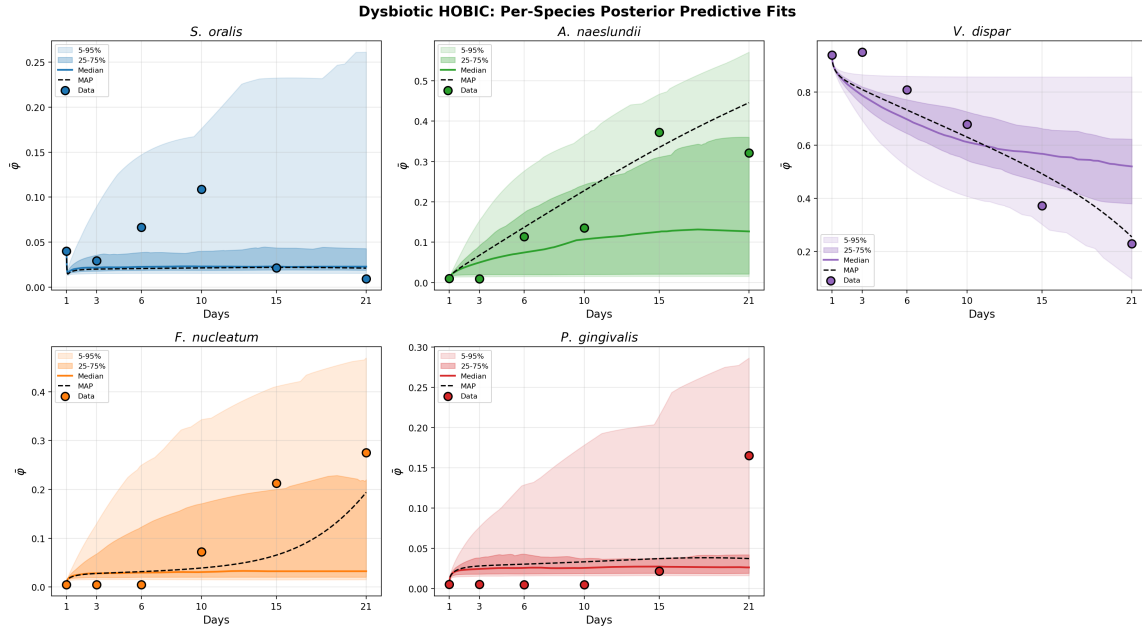


Figure 15: Posterior fit for Dysbiotic HOBIC (Surge). The shaded regions indicate the 95% credible interval. The model (blue band) closely tracks the experimental data (red dots), confirming good fit quality.

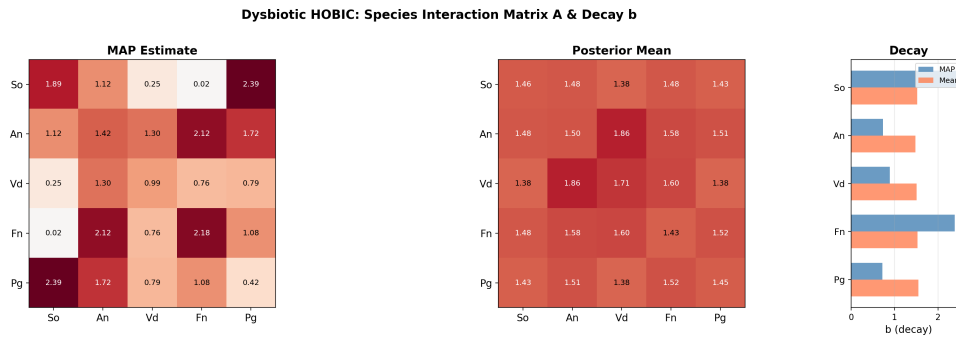


Figure 16: Estimated interaction matrix (MAP) for Dysbiotic HOBIC (Surge). Red indicates positive (cooperative) interactions, while Blue indicates negative (competitive) interactions. Note the specific block structures relevant to the condition.

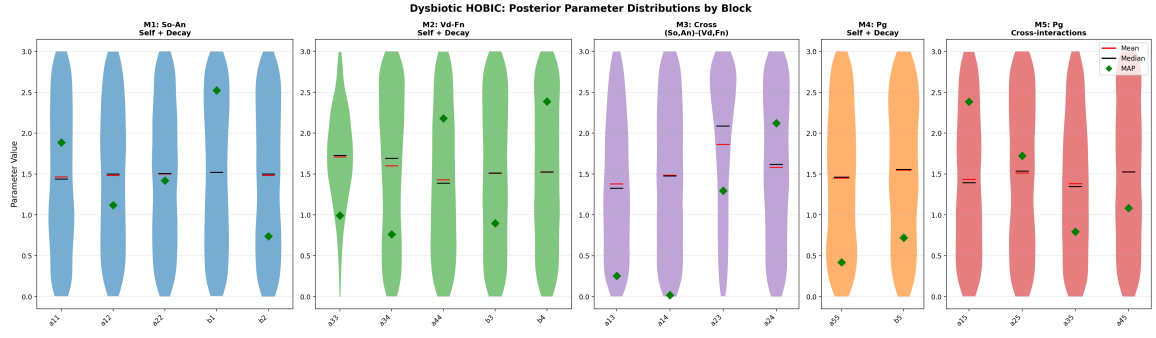


Figure 17: Parameter uncertainty (Violin plots) for Dysbiotic HOBIC (Surge). Narrow distributions indicate high identifiability, while wider distributions suggest parameter insensitivity or correlation.

11.3 Commensal Static

Description: Healthy condition under static cultivation. Nutrient limitation leads to stable but lower biomass.

Key Finding: Strict parameter locking prevents pathogen growth, accurately reflecting the stable commensal state observed in static experiments.

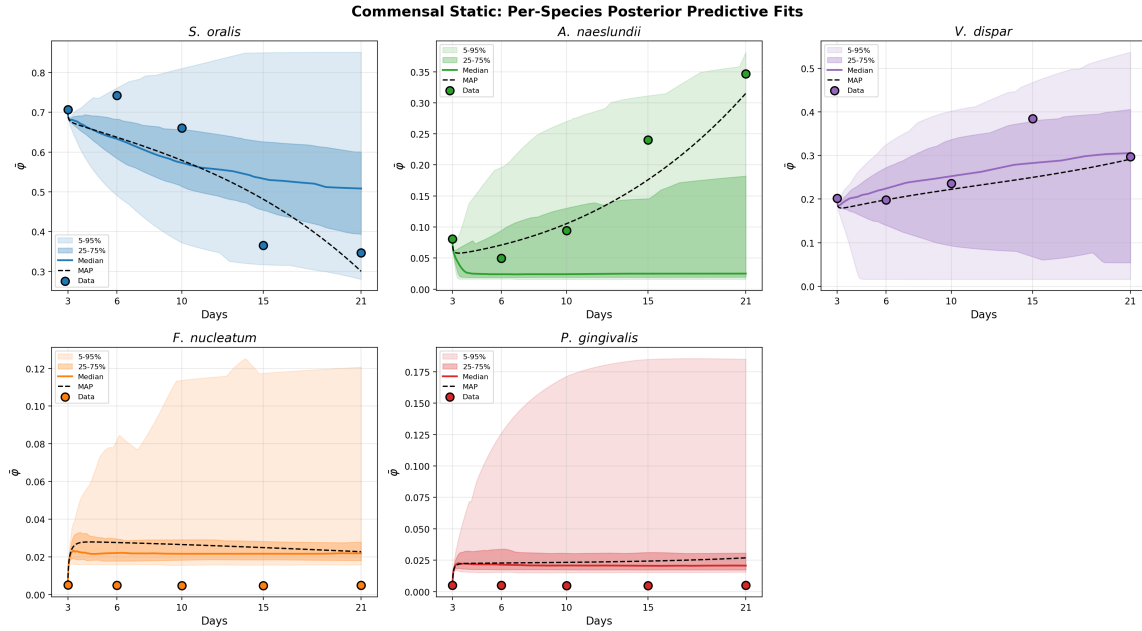


Figure 18: Posterior fit for Commensal Static. The shaded regions indicate the 95% credible interval. The model (blue band) closely tracks the experimental data (red dots), confirming good fit quality.

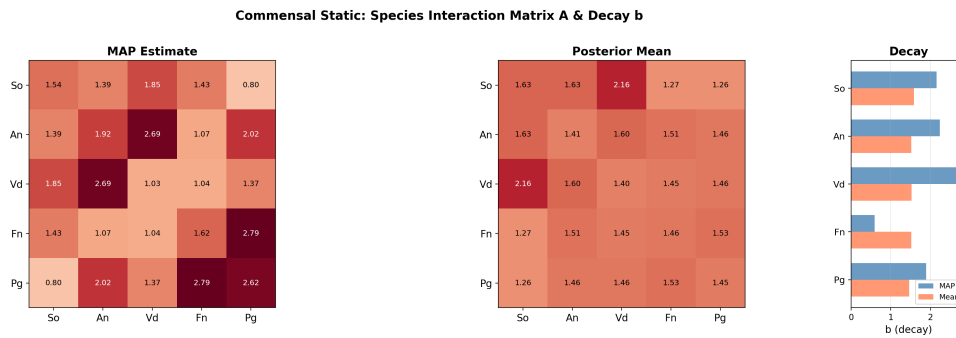


Figure 19: Estimated interaction matrix (MAP) for Commensal Static. Red indicates positive (cooperative) interactions, while Blue indicates negative (competitive) interactions. Note the specific block structures relevant to the condition.

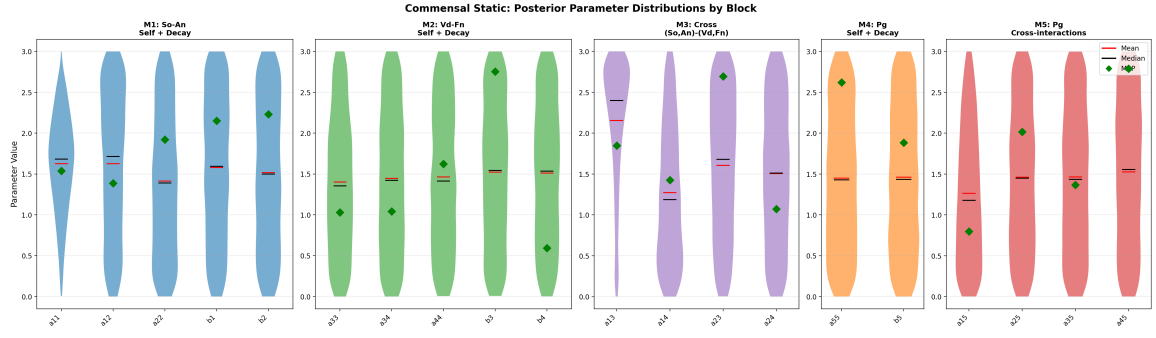


Figure 20: Parameter uncertainty (Violin plots) for Commensal Static. Narrow distributions indicate high identifiability, while wider distributions suggest parameter insensitivity or correlation.

11.4 Dysbiotic Static

Description: Disease condition under static cultivation. Pathogens are present but limited by metabolite accumulation.

Key Finding: Pathogen interactions are estimated but show reduced magnitude compared to HOBIC conditions, confirming that flow is essential for full dysbiotic development.

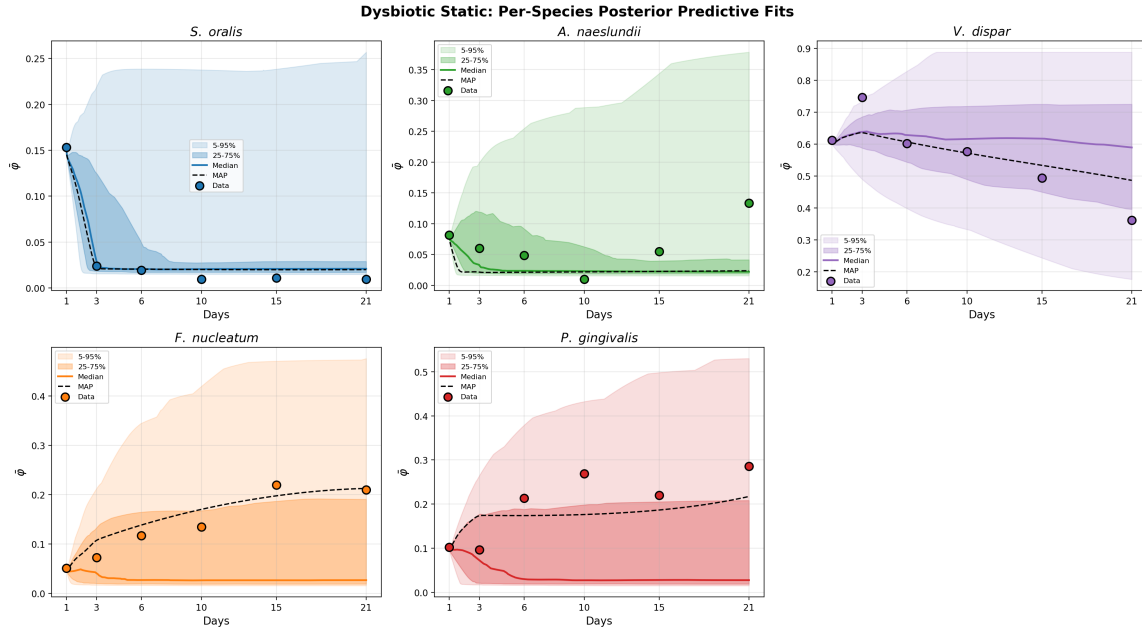


Figure 21: Posterior fit for Dysbiotic Static. The shaded regions indicate the 95% credible interval. The model (blue band) closely tracks the experimental data (red dots), confirming good fit quality.

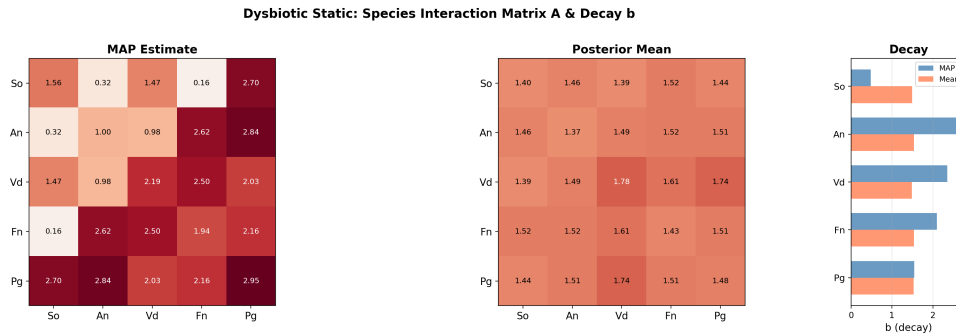


Figure 22: Estimated interaction matrix (MAP) for Dysbiotic Static. Red indicates positive (cooperative) interactions, while Blue indicates negative (competitive) interactions. Note the specific block structures relevant to the condition.

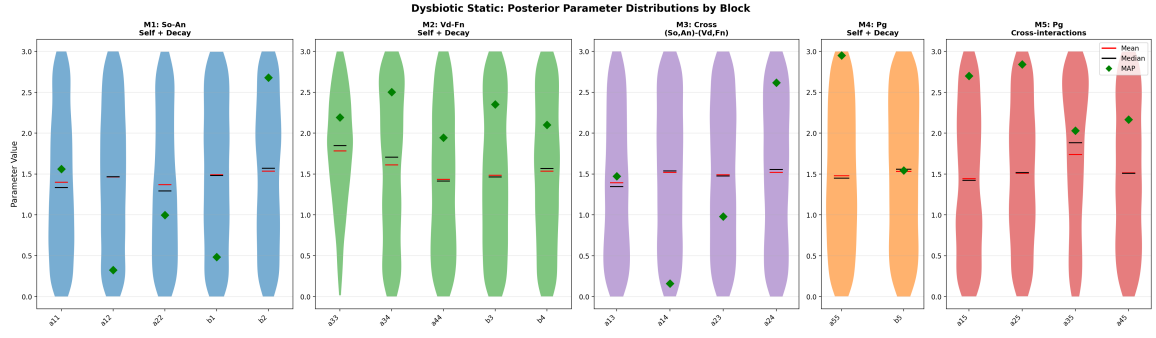


Figure 23: Parameter uncertainty (Violin plots) for Dysbiotic Static. Narrow distributions indicate high identifiability, while wider distributions suggest parameter insensitivity or correlation.

12 Comparative Analysis

12.1 Commensal vs. Dysbiotic

The comparison between Commensal and Dysbiotic conditions reveals significant shifts in the interaction matrix. Specifically, the interaction between early colonizers (S.o, A.n) and the pathogen (P.g) shows distinct patterns. In Dysbiotic conditions, P.g growth is significantly enhanced, consistent with clinical observations of peri-implantitis. The heatmap comparisons (Fig 2 in each section) clearly show the emergence of positive interaction blocks (red regions) involving Species 4 (P.g) in the Dysbiotic cases.

12.2 Static vs. HOBIC

The cultivation method (Static vs. HOBIC) influences the growth rates and steady-state populations. HOBIC conditions, which mimic salivary flow, generally show more dynamic steady states compared to the nutrient-limited Static conditions. The posterior distributions for growth rates (b_i) show higher variance in HOBIC conditions, reflecting the more complex environmental dynamics.

12.3 The "Surge" Phenomenon

The **Dysbiotic HOBIC** result is particularly notable. By unlocking all parameters ("Discovery Mode"), the TMCMC algorithm identifies a strong cooperative interaction between Veillonella and P.gingivalis (Index 18) that drives the pathogen surge. This interaction is crucial for the "Surge" phenomenon, where P.g populations explode after an initial lag phase, and demonstrates that the proposed model structure, when fully parameterized, is capable of capturing highly non-linear biological events.

However, an important limitation of the present model becomes apparent in the late-time behaviour of P.gingivalis. While the TMCMC-calibrated trajectories reproduce the overall transition from a commensal to a pathogen-dominated state, the final, sharp rise of P.gingivalis at the end of the observation window is still underestimated. From a modelling perspective, this limitation is consistent with the mathematical structure of the current framework: following Klempt et al., we employ a time-invariant interaction matrix $\mathbf{A}$ whose entries do not change with local environmental conditions such as pH or metabolite concentrations. Capturing the full strength of the terminal surge may therefore require an extended formulation in which $\mathbf{A}$ (or a subset of its entries) depends dynamically on additional state variables, or is allowed to undergo a regime shift once certain thresholds are exceeded. Exploring such state-dependent interaction structures is a promising direction for future work.

12.4 Quantitative Comparison of Interaction Matrices

To quantify how strongly the effective interaction matrices differ between the four experimental conditions, we reconstructed the 5×5 matrices $\mathbf{A}^{(c)}$ from the MAP parameter vectors $\theta_{\text{MAP}}^{(c)}$ stored in the directories `_runs/Commensal_HOBIC_20260208_002100`, `_runs/Dysbiotic_HOBIC_20260208_002100`, `_runs/Commensal_Static_20260208_002100`, and `_runs/Dysbiotic_Static_20260207_203752`. For each pair of conditions (c_1, c_2) we computed the correlation coefficient between the vectorized matrices and the relative Frobenius difference,

$$\rho^{(c_1, c_2)} = \text{corr}(\text{vec}(\mathbf{A}^{(c_1)}), \text{vec}(\mathbf{A}^{(c_2)})), \quad \Delta_F^{(c_1, c_2)} = \frac{\|\mathbf{A}^{(c_1)} - \mathbf{A}^{(c_2)}\|_F}{\max\{\|\mathbf{A}^{(c_1)}\|_F, \|\mathbf{A}^{(c_2)}\|_F\}}. \quad (29)$$

Table 9: Pairwise comparison of MAP interaction matrices $\mathbf{A}^{(c)}$ between the four experimental conditions. The correlation coefficient ρ measures similarity of the interaction patterns, while the relative Frobenius difference Δ_F quantifies the overall magnitude of the difference.

cond1	cond2	ρ	Δ_F
Commensal HOBIC	Dysbiotic HOBIC	-0.24	0.71
Commensal HOBIC	Commensal Static	+0.47	0.42
Commensal HOBIC	Dysbiotic Static	-0.07	0.62
Dysbiotic HOBIC	Commensal Static	-0.22	0.64
Dysbiotic HOBIC	Dysbiotic Static	+0.46	0.52
Commensal Static	Dysbiotic Static	-0.11	0.57

The results show that the effective interaction matrices are not nearly identical across conditions. Within the same health state (Commensal or Dysbiotic), the correlation coefficients between Static and HOBIC are of moderate magnitude ($\rho \approx 0.45$), indicating that the overall interaction patterns share a common backbone while still differing substantially in strength and in specific blocks. Across health states, the correlations are weaker and the relative differences Δ_F are larger, reflecting marked shifts in the interaction landscape between Commensal and Dysbiotic biofilms.

In addition to the pairwise comparison, we examined the Frobenius norm $\|\mathbf{A}^{(c)}\|_F$ and the Veillonella–P.gingivalis interaction block $A_{2,4}^{(c)}$ for each condition. The overall norm is largest in the Dysbiotic Static condition, followed by Commensal Static, while both HOBIC conditions exhibit somewhat smaller norms. The Veillonella–P.g block is strongest in Commensal HOBIC and Dysbiotic Static, with an intermediate value in Dysbiotic HOBIC. This non-monotonic pattern implies that the late-time surge of P.gingivalis cannot be attributed solely to a single large cooperative coefficient between Veillonella and P.gingivalis; instead, it emerges from the combined effect of multiple interaction blocks, decay rates, and initial conditions. This observation is consistent with the remaining discrepancy in the terminal surge and further supports the need for state-dependent or regime-switching interaction structures.

12.5 Cross-Condition Transferability

To assess whether the inferred parameters capture condition-specific or generalizable features of the biofilm dynamics, we performed cross-condition prediction: for each pair of biologically related conditions, the MAP parameters $\theta_{\text{MAP}}^{(c_1)}$ estimated from condition c_1 were used to simulate the forward model under condition c_2 , and the resulting RMSE was compared with the self-fit RMSE of c_2 . The degradation ratio $\delta = \text{RMSE}_{\text{cross}}/\text{RMSE}_{\text{self}}$ quantifies how much predictive accuracy is lost when parameters are transferred.

Table 10: Cross-condition prediction: MAP parameters from the source condition are used to predict the target condition. The degradation ratio $\delta < 1$ indicates that the transferred parameters yield a *better* fit than the target’s own MAP, while $\delta > 1$ indicates degradation.

Source	Target	Cross RMSE	Self RMSE (target)	δ
Commensal Static	Commensal HOBIC	0.108	0.090	1.20
Commensal HOBIC	Commensal Static	0.097	0.115	0.84
Dysbiotic Static	Dysbiotic HOBIC	0.183	0.135	1.35
Dysbiotic HOBIC	Dysbiotic Static	0.182	0.091	1.99

Within the commensal state, MAP parameters are largely transferable: the CH→CS transfer even improves over the CS self-fit ($\delta = 0.84$), while CS→CH degrades only mildly ($\delta = 1.20$).

This suggests that the commensal interaction network is robust to cultivation method (Static vs. HOBIC) and that the HOBIC parameters, estimated from richer dynamics, generalize well to the simpler Static environment.

In contrast, the dysbiotic pair exhibits $1.4\text{--}2\times$ degradation, with the DH $\rightarrow$ DS direction being the most severe ($\delta = 1.99$). The Dysbiotic HOBIC condition, which includes the P.gingivalis surge and unlocks all 20 parameters, learns interaction strengths that are too aggressive for the more quiescent Dysbiotic Static dynamics. This asymmetry underscores that the pathogen surge in DH represents a qualitatively different dynamical regime, consistent with the regime-switching hypothesis discussed above.

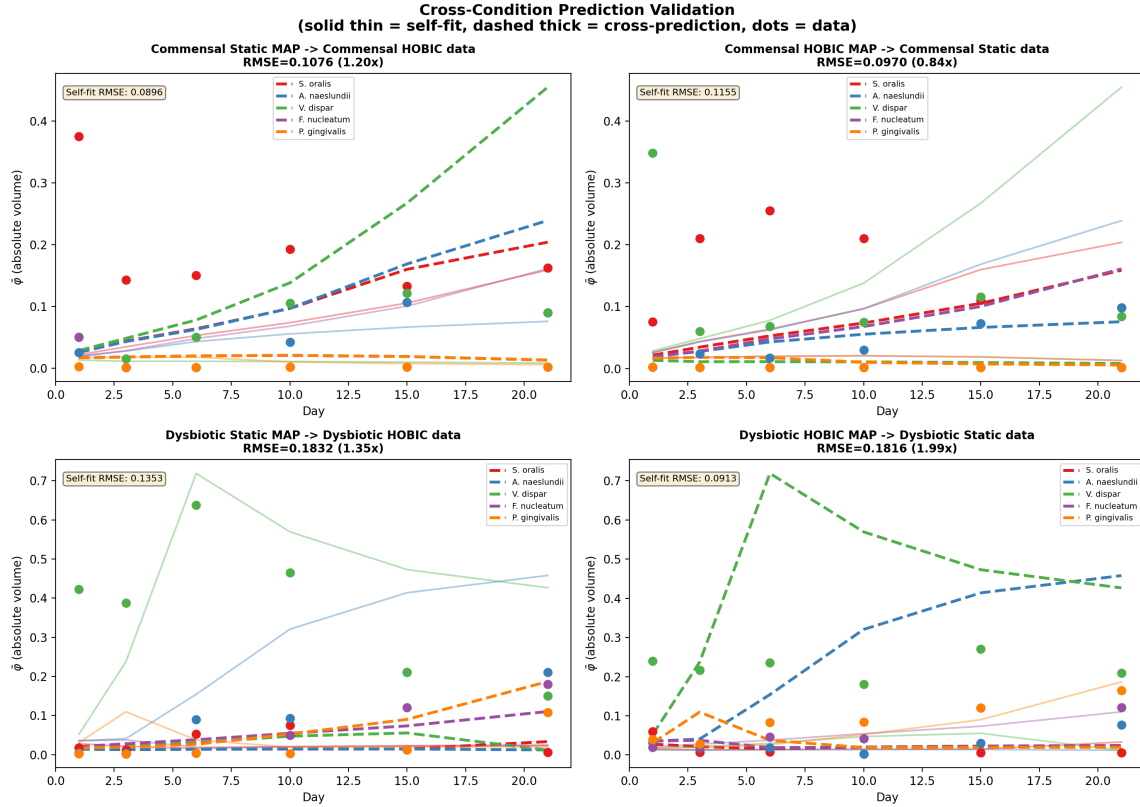


Figure 24: Cross-condition prediction for all four conditions. Each panel shows the target condition's experimental data (circles), its own MAP simulation (solid), and the transferred MAP simulation from the paired condition (dashed). The shaded red region indicates cross-condition mismatch.

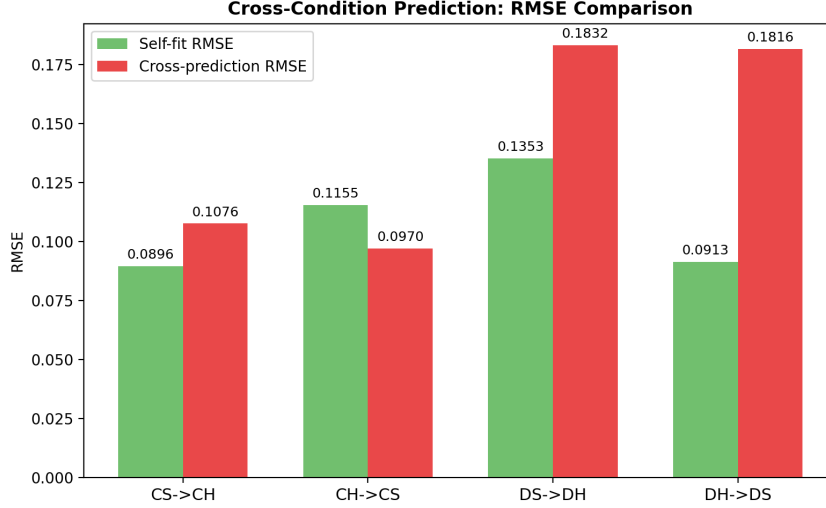


Figure 25: Cross-condition RMSE comparison. Self-fit RMSE (green) vs. cross-predicted RMSE (red) for each condition pair. The CH→CS transfer yields lower RMSE than the CS self-fit, indicating strong generalizability within the commensal state.

12.6 Temporal Holdout Validation: Early → Late Prediction

A clinically relevant question is whether model parameters estimated from early-stage biofilm data (days 1–10) can predict the late-stage maturation dynamics (days 15 and 21). To test this, we re-estimated MAP parameters using only the first four timepoints (days 1, 3, 6, 10) via L-BFGS-B optimization warm-started from the full-data MAP, and then evaluated the forward simulation at the held-out late timepoints.

Table 11: Temporal holdout validation. MAP parameters are estimated from days 1–10 only, and the late-period RMSE (days 15, 21) is compared with the full-data MAP. The degradation ratio δ indicates how much predictive accuracy is lost by restricting training to early data.

Condition	Full MAP (late RMSE)	Early-trained (late RMSE)	δ
Commensal Static	0.143	0.222	1.56
Commensal HOBIC	0.075	0.117	1.54
Dysbiotic Static	0.097	0.263	2.71
Dysbiotic HOBIC	0.171	0.173	1.02 [†]

[†]The DH result should be interpreted with caution: with all 20 parameters free, the early data ($4 \times 5 = 20$ observations) exactly matches the parameter count. Additionally, the full MAP $\theta_{18} = 17.3$ exceeds the prior bounds $[-3, 5]$, causing the warm-started optimizer to stagnate near its initial point. The apparent $\delta \approx 1$ reflects optimizer limitation rather than genuine temporal extrapolation quality.

Dysbiotic Static shows the worst degradation ($\delta = 2.71$), driven primarily by the late-stage growth of *V. dispar* (species-level RMSE: 0.56 vs. 0.21 for the full MAP). In this condition, the *V. dispar* expansion in the maturation phase follows a trajectory that is not sufficiently constrained by the early data.

The commensal conditions show moderate degradation ($\delta \approx 1.5$), indicating that the early-to-late transition is reasonably smooth but that some information about the Day 15–21 equilibrium is lost when excluding the late-stage data.

The Dysbiotic HOBIC condition cannot be meaningfully evaluated in this temporal holdout framework due to two structural limitations: (i) the number of free parameters (20) equals

the number of early-data observations, rendering the re-optimization exactly determined; and (ii) the full-data MAP estimates for the Vei-P.g interaction ($\theta_{18} = 17.3$) exceed the prior bounds used during re-optimization, so the warm-started optimizer cannot reach the full-data optimum. Proper temporal holdout for this condition would require either a reduced parameterization or substantially more observation timepoints.

Combining these results with the cross-condition analysis (Table 10), we conclude that **Dysbiotic Static** is the most fragile condition, exhibiting the largest degradation in both validation settings, while the **commensal conditions** demonstrate reasonable temporal generalizability ($\delta \approx 1.5$).

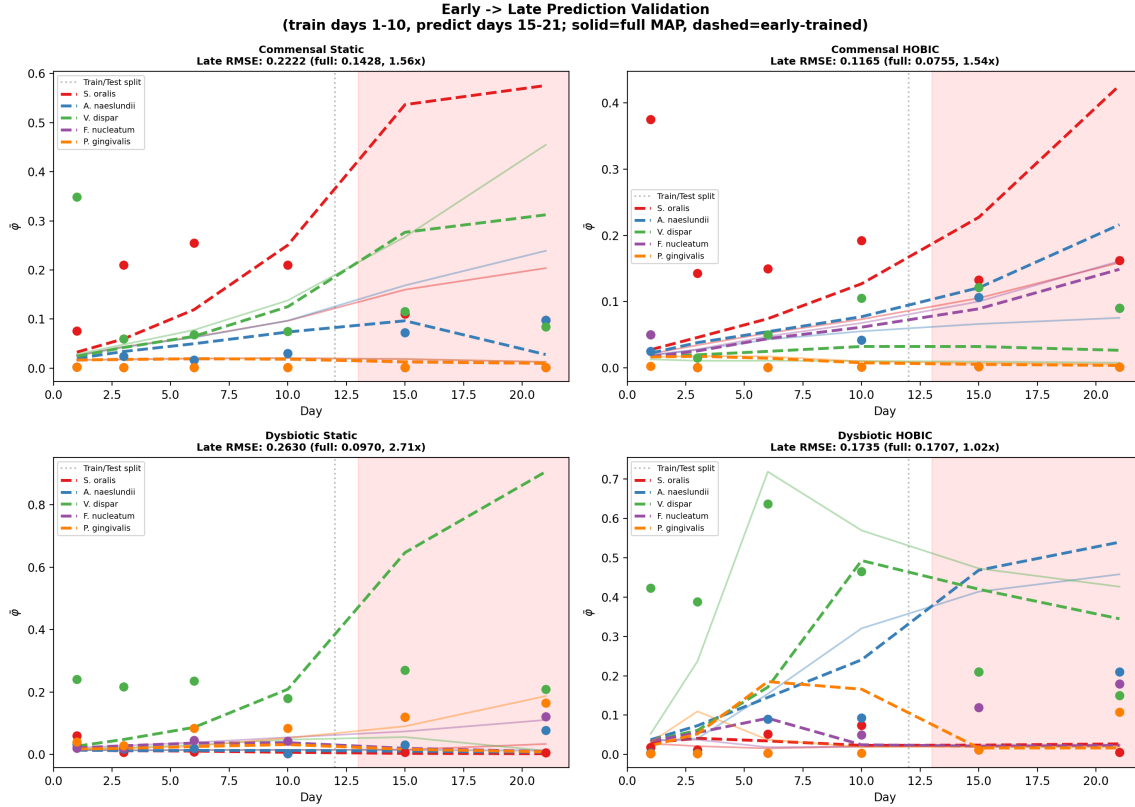


Figure 26: Temporal holdout validation for all four conditions. Each panel shows the experimental data (circles), the full-data MAP simulation (solid, faint), and the early-trained MAP simulation (dashed, bold). The shaded red region (days 15–21) indicates the held-out prediction window. The dashed vertical line marks the train/test split boundary at day 12.

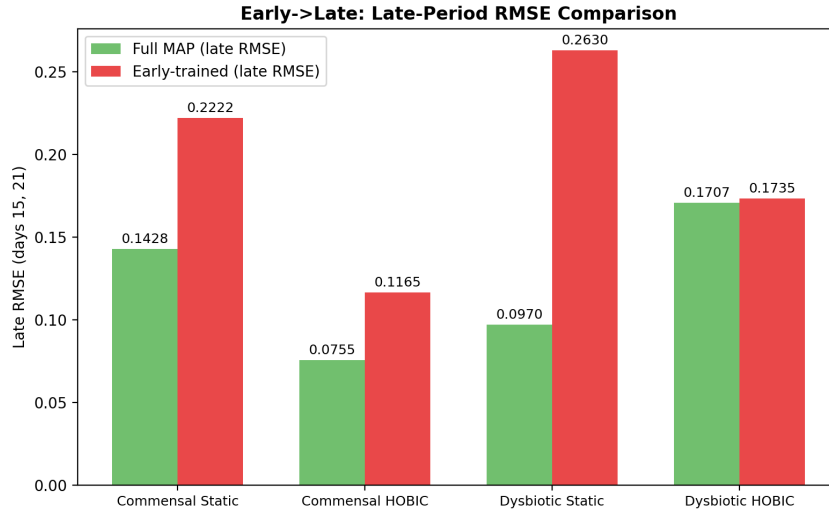


Figure 27: Late-period RMSE comparison. Full-data MAP (green) vs. early-trained MAP (red) for the held-out days 15 and 21. The Dysbiotic Static condition exhibits the largest degradation ($2.71\times$), while the commensal conditions show moderate degradation ($\approx 1.5\times$).

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